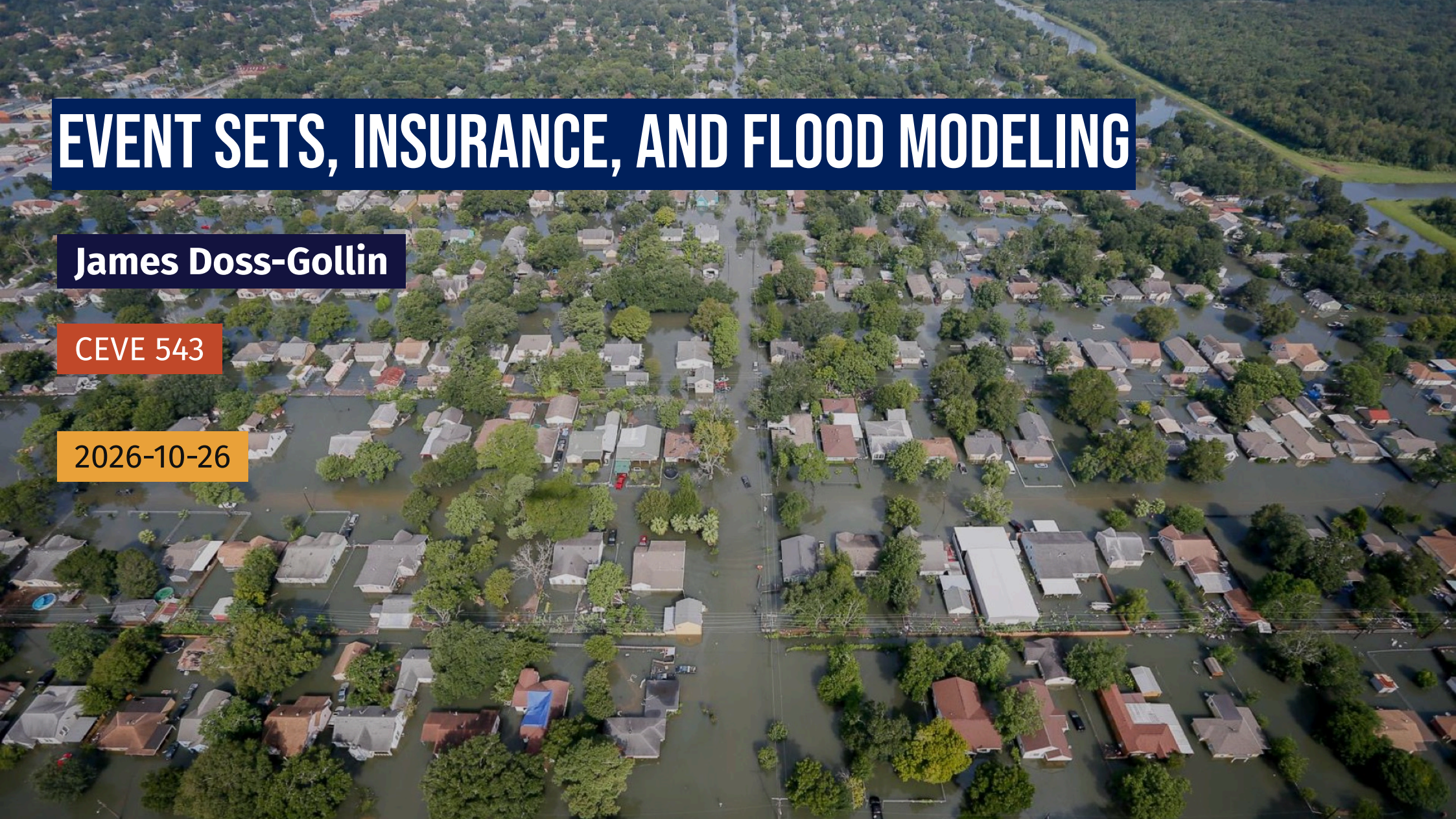
EVENT SETS, INSURANCE, AND FLOOD MODELING

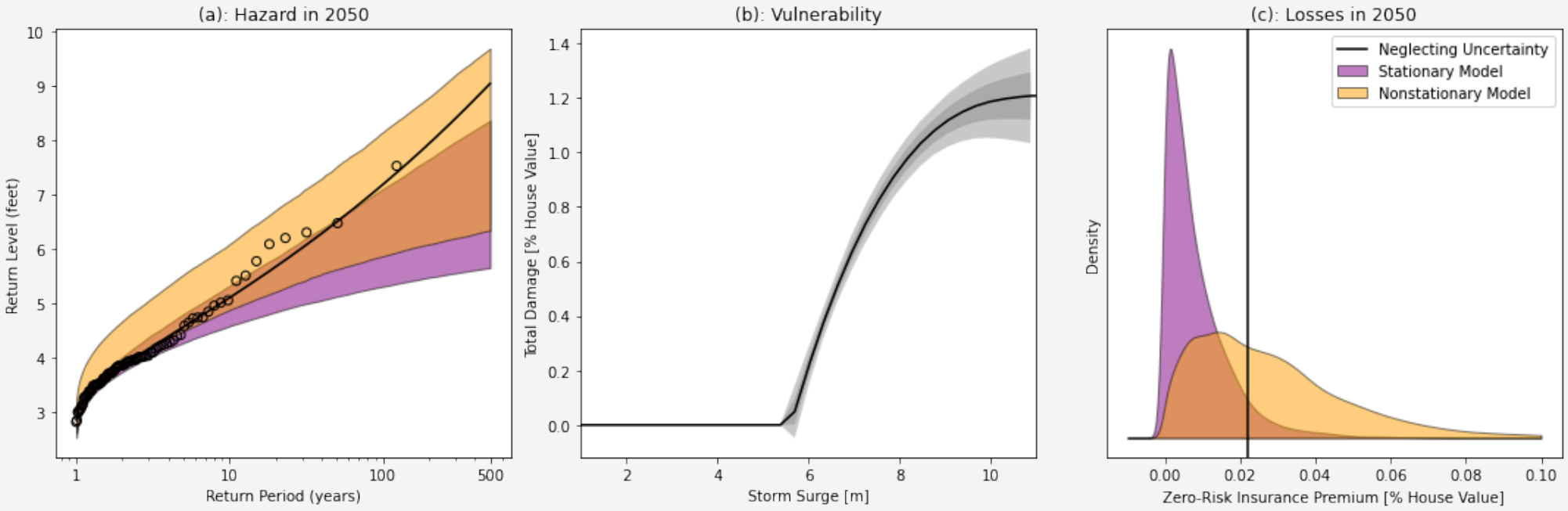
James Doss-Gollin

CEVE 543

2026-10-26



WHAT IS THE DISTRIBUTION OF FLOOD LOSSES IN A CITY?



Flood hazard, house vulnerability, and the resulting distribution of insurance premiums. Own work.

COULD WE FIT A GEV OR GPD TO PAST FLOOD LOSSES?

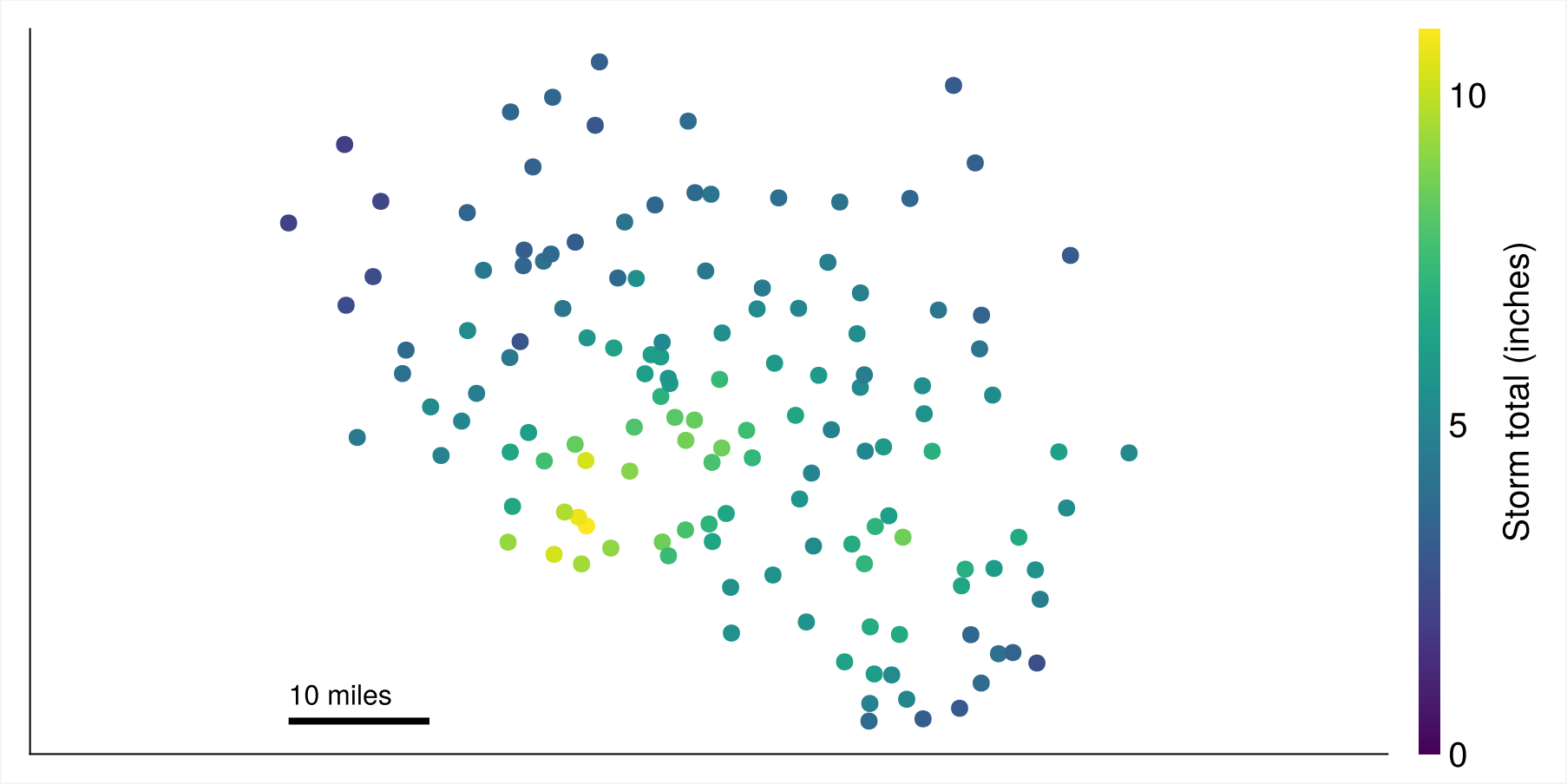


A submerged suburban neighborhood after Hurricane Harvey. Photo by Staff Sergeant Martinez, public domain.

AN EVENT SET: REALISTIC EVENTS WE CAN COMPUTE DAMAGE FROM

Event	Rate (per year)	Loss (million dollars)
1	0.002	410
2	0.005	160
3	0.010	95
4	0.040	22
⋮	⋮	⋮

MEMORIAL DAY 2015: WHERE DID THE RAIN FALL?



Totals for May 25-26 at Harris County Flood Control District gauges.

WHERE DID MEMORIAL DAY 2015 FLOOD BUILDINGS?

i Note

Figure shown in class: Buildings flooded in 2015 and 2016, many outside FEMA's high-risk zones. Satija et al. (2016).

What was the return period of this flood, and how would it enter average annual loss?

WHERE COULD HURRICANES HAVE GONE THAT THE RECORD HAS NOT SEEN?

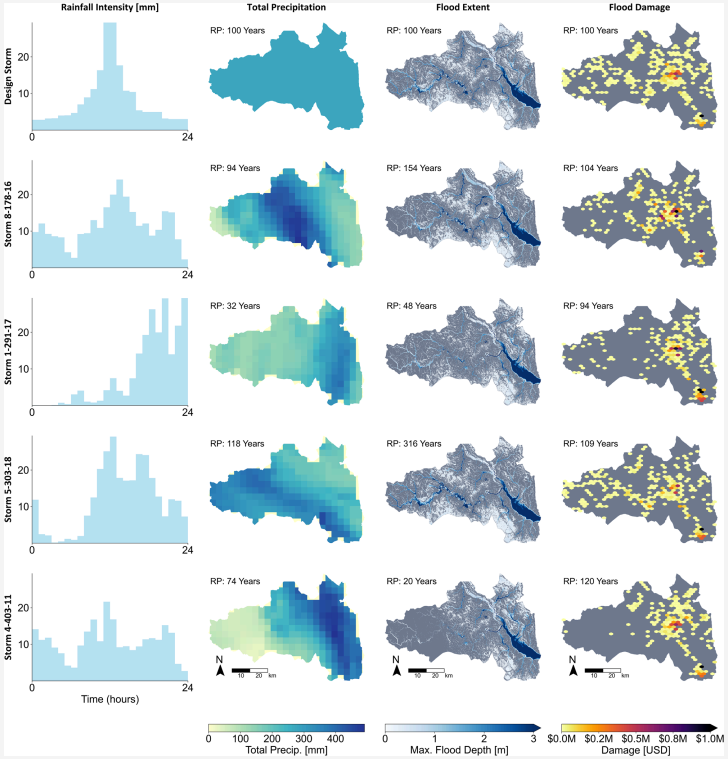
Note

Figure shown in class: Historical major hurricane tracks beside simulated tracks from the same coast. O'Rourke & Burdick (2025), p. 5.

WHERE DO REALISTIC EVENTS COME FROM?

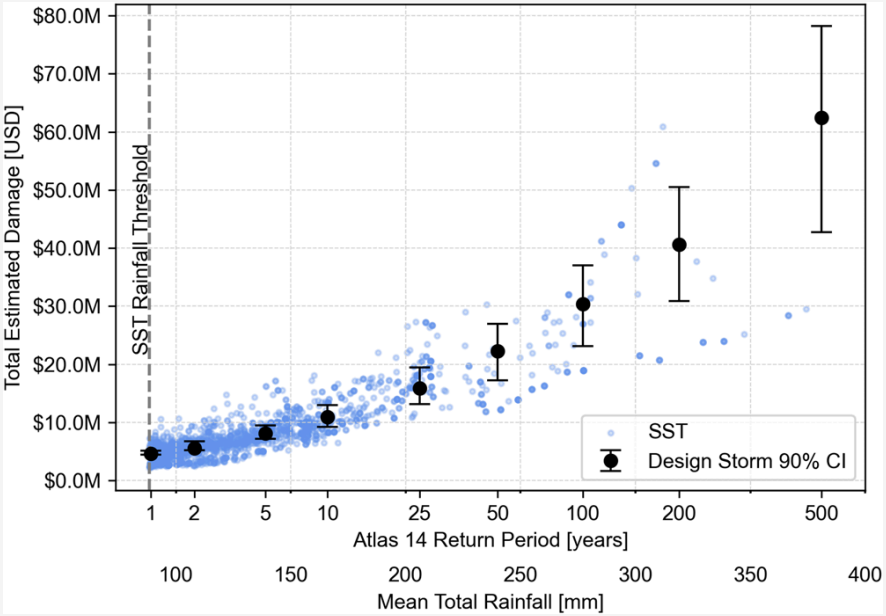
Storm transposition	Climate models	Weather generators
This week	Week 11	Week 13

DOES IT MATTER HOW THE STORMS ARE BUILT?



Four transposed storms with about 100-year damage, and the design storm. Baer et al. (2026), Fig. 4.

THE SAME RAINFALL?



Damage against basin rainfall, transposition and design storms. Baer et al. (2026), Fig. 2.

Expected annual damage: 14.2 million dollars a year from transposition, 7.4 million more than from design storms (Baer et al., 2026, p. 3).